

Report

Nature of Data

The above data consisted of an output from an electrostatic lens under the influence of a field consisting of mainly 200 particles comprising of its position followed by the Cartesian coordinates traced by the left hand rule and the voltage for each particles with its corresponding Energy and Aberration coefficient for the same and the trajectory spread .

Methodology: The notebook that I have used here is Google colab and various libraries for plotting and regression

1) Cleaning and preprocessing data :

During cleaning data I have found that there were 5 missing data points that is the velocity of particles and 12 different outliers which I have indicated using basic commands and reproduced the missing velocities by a median fill command the 12 outliers which were extreme values from both have been omitted for further analysis as it would only make the data more complex

2) EDA and Relevant visualizations:

Here for visualization I have used the inbuilt libraries such as matplotlib for various scatter and line plots I have plotted the histograms for velocity, spherical aberration and Energy in all of those three plots I have found a close to a normal distribution. I have plotted the scatter plot for velocity vs Energy, voltage vs Spherical aberration and voltage vs velocity among all three the relation between energy and velocity has been a direct relation as there is a one to one increase in energy with velocity with other two plots indicating a less intuitive nature of data

3) Correlation Analysis and Regression

The correlation map is showing the correlation between variables here in this case the only strong correlation can be seen is between Velocity_mps and Energy_joules also I mentioned earlier from the graph also followed by the least Correlation between Voltage and spherical aberration and almost zero correlation between voltage and energy

4) Machine Learning for optimizing lens parameters

The effective optimization of lens parameters such as spherical aberration coefficient and spread and potential arrays can be predicted on the basis of dynamic artificial neural networks to predict the voltage ratios for electrodes with minimum aberrations which has a feedback communication which are compatible with time series as ANN focuses on predicting time series data which consist of 2 input neuron and 1 output neuron and a hidden layer.

Findings:

- 1) From the above data and the results that I got from the correlation between the parameters I found that that the lens is not effectively accelerating particles as there is a little to no correlation between voltage and aberration and spread values
- 2) A regression model between spherical aberration and all the other variables results in a very low spread and coefficient which means spread slightly increases with aberration but overall effect is minimum here.

Suggestions:

Potential arrays are very important in order to communicate an electrostatic lens which tells a lot about the geometry of lenses as it calculates the field at every point from where the particle is accelerated and which could be the key to even predicting or training several machine learning models for better lens parameter optimization as from what I've known best about the Electrostatic lenses you can import the geometry of lenses using CAD files or STL or TXT formats taking that into account is important .